# DA2402 Data Curation and Visualisation Project

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## Data Dictionary

The table below documents the columns of the cleaned dataset `budget_nfhs_combined_clean.csv`, grouped by category. Most NFHS-5 indicators repeat in Urban, Rural and Total form; only the variants confirmed in the cleaning notebook are listed individually below. Remaining indicator columns follow the same naming, dtype, unit and range convention and should be added as further rows directly from the working dataframe's column list.

| Column | Dtype | Unit | Source | Allowed Range |
|---|---|---|---|---|
| **Identifiers** | | | | |
| State | object | n/a | NFHS-5 / RBI / NHM | 29 Indian states |
| **Population and Expenditure (raw)** | | | | |
| Population (Total) | float64 | persons | NHM Population Projections | $> 0$ |
| Accounts | float64 | Rs. Lakh | RBI State Finances | $\geq 0$ |
| Budget Estimates | float64 | Rs. Lakh | RBI State Finances | $\geq 0$ |
| Revised Estimates | float64 | Rs. Lakh | RBI State Finances | $\geq 0$ |
| **Expenditure (derived)** | | | | |
| Per_Capita_Health_Exp_Accounts | float64 | Rs. per person | Derived (Accounts / Population) | $\geq 0$ |
| Per_Capita_Health_Exp_BE | float64 | Rs. per person | Derived (Budget Estimates / Population) | $\geq 0$ |
| Per_Capita_Health_Exp_RE | float64 | Rs. per person | Derived (Revised Estimates / Population) | $\geq 0$ |
| **Demographic Context** | | | | |

| Column | Dtype | Unit | Source | Allowed Range |
|---|---|---|---|---|
| Population below age 15 years (%) (Urban/Rural/Total) | float64 | % | NFHS-5 | $[0, 100]$ |
| Sex ratio of the total population (females per 1,000 males) | float64 | ratio | NFHS-5 | $> 0$ |
| Sex ratio at birth for children born in the last five years | float64 | ratio | NFHS-5 | $> 0$ |
| **Mortality** | | | | |
| Neonatal mortality rate (NNMR) (Urban/Rural/Total) | float64 | per 1,000 live births | NFHS-5 | $\geq 0$ |
| Infant mortality rate (IMR) (Urban/Rural/Total) | float64 | per 1,000 live births | NFHS-5 | $\geq 0$ |
| Under-five mortality rate (U5MR) (Urban/Rural/Total) | float64 | per 1,000 live births | NFHS-5 | $\geq 0$ |
| **Birth Registration and Early Childhood Services** | | | | |
| Children under age 5 years whose birth was registered (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Children age 5 years who attended pre-primary school during 2019-20 (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Women age 15-19 years who were already mothers (%) (Urban/Rural) | float64 | % | NFHS-5 | $[0, 100]$ |
| **Maternal and Delivery Care** | | | | |
| Institutional births (%) (Urban/Rural/Total) | float64 | % | NFHS-5 | $[0, 100]$ |
| Children born at home taken to a health facility for a check-up within 24 hours (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Mothers whose last birth was protected against tetanus (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| **Child Feeding and Nutrition** | | | | |
| Children under age 6 months exclusively breastfed (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Children age 6-8 months receiving solid/semi-solid food and breastmilk (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Non-breastfeeding children age 6-23 months receiving an adequate diet (%) | float64 | % | NFHS-5 | $[0, 100]$ |

| Column | Dtype | Unit | Source | Allowed Range |
|---|---|---|---|---|
| Total children age 6-23 months receiving an adequate diet (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Children under 5 years who are stunted (height-for-age) (%) (Urban/Rural/Total) | float64 | % | NFHS-5 | $[0, 100]$ |
| Children under 5 years who are overweight (weight-for-height) (%) (Urban/Rural) | float64 | % | NFHS-5 | $[0, 100]$ |
| **Immunisation** | | | | |
| Children age 12-23 months who have received BCG (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Children age 12-23 months fully vaccinated, card or mother's recall (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Children age 12-23 months fully vaccinated, card only (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Children age 12-23 months who received most of their vaccinations (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Children age 12-23 months who received the first dose of MCV (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| Children age 24-35 months who received a second dose of MCV (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| **Health Financing** | | | | |
| Average out-of-pocket expenditure per delivery in a public facility | float64 | Rs. | NFHS-5 | $\geq 0$ |
| **Anaemia** | | | | |
| Pregnant women age 15-49 years who are anaemic (%) | float64 | % | NFHS-5 | $[0, 100]$ |
| **Women's Health and Education** | | | | |
| Women whose Body Mass Index (BMI) is below normal (%) (Urban/Rural/Total) | float64 | % | NFHS-5 | $[0, 100]$ |
| Women with 10 or more years of schooling (%) (Urban/Rural/Total) | float64 | % | NFHS-5 | $[0, 100]$ |
| **Household Infrastructure** | | | | |

| Column | Dtype | Unit | Source | Allowed Range |
|---|---|---|---|---|
| Population living in households with an improved drinking water source (%) | float64 | % | NFHS-5 | $[0, 100]$ |

Table 1: Data dictionary for the cleaned dataset, grouped by category. Remaining NFHS-5 indicator columns follow the same Urban/Rural/Total and percentage convention.